

Rafi Khan

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Education

The Ohio State University

Master of Science in Mechanical Engineering (Thesis), GPA: 3.94

Expected December 2026

Bachelor of Science in Mechanical Engineering, GPA: 3.68

December 2024

Experience

BWX Technologies, Barberton, OH

May 2026 – August 2026

Design Stress Engineering Intern

- Performed FEA thermal stress relief analysis on a naval reactor heat exchanger per NAVSEA structural requirements
- Conducted quench analysis on a forged steel component and performed brittle fracture hand calculations from the results to demonstrate fracture resistance per NAVSEA requirements

Materials at Extremes Lab, OSU

Jan 2025 – Present

Graduate Research Assistant

- Developed an iterative creep model framework to predict strain and rupture of Ni-base superalloys from partial data
- Implemented in MATLAB (Optimization Toolbox, Ohio Supercomputer); validated on code-qualified Alloy 617, reaching converged creep-life predictions from ~25% of test duration
- Published in ASME *J. Eng. for Gas Turbines & Power* (GTP-26-1405) and presented at ASME Turbo Expo 2026, Milan
- Modeled novel accelerated creep test (Delta-T DIC) that extracts hundreds of ASTM-standard creep curves across a continuous temperature gradient from a single specimen

SpaceX, Hawthorne, CA

May 2023 – August 2023

Supplier Development Engineering Intern, Starlink Aviation

- Implemented high voltage testing of Starlink harnesses for the first time in the department. Owned all the processes from test hardware setup, fabrication, programming, and implementation in the reliability lab.
- Uncovered novel defects when performing tests per Federal Aviation Administration (FAA) requirements
- Led the manufacturing qualification for a new Starlink Aviation product, identifying critical gaps in the supplier's process and correcting their tensile-strength test procedures to achieve IPC-620 compliance
- Identified a non-conformance (NC) on over 3000 in-stock harnesses planned for installation on Airbus A320s, placing NCs into containment and passing FAA certification

General Electric Aerospace, Dayton, OH

September 2022 – December 2022

Manufacturing Engineering Intern, Aviation Systems

- Assembled high-voltage (1500 V) power converters for a new hybrid aircraft program by interpreting complex engineering drawings and design requirements
- Authored detailed work instructions based on the initial build process, creating a repeatable standard that streamlined assembly for technicians on future units
- Designed and fabricated a custom screen-printing jig to apply thermal paste onto sensitive semiconductor modules, establishing a repeatable process to ensure reliable thermal management
- Acquired and applied high-reliability skills by fabricating aerospace-grade wire harnesses, terminating fiber optic cables, and performing certified soldering according to IPC-610 standards

Denso, Maryville, TN

May 2022 – August 2022

Process Design Intern

- Optimized a high-volume automated production line for computer modules by using Cpk analysis to identify packaging flaws and establishing a new box orientation procedure that eliminated line crashes

Skills

SolidWorks (Certified), MATLAB, Ansys, Abaqus, HSMWorks (CAM), Python, 3D-Printing